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The education-related digital divide: An analysis for the EU-28

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ABSTRACT

This paper addresses the international and internal digital divides that exist across and within the European member states according to the educational attainment of their populations. Our results suggest that even for those European countries that are outperforming their counterparts in terms of digital development, such as Finland, some internal gaps still remain and need to be addressed. In other countries, as in the cases of Malta, Spain, and Portugal, the divides are a matter for concern. These findings would probably be overlooked if we worked only with aggregate levels, as is usual. Consequently, this paper draws attention to the importance of complementing cross-country analysis of the digital divide with an assessment of internal gaps.

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1. Introduction

Although information and communication technologies (ICT) are today profoundly intertwined with almost every aspect of economic and social activities, they still continue to hold the promise of tremendous innovation and development opportunities, provided that the right enabling conditions are put into place (European Commission, 2013). The conviction that greater adoption and use of ICT will foster growth and development, trumping the present economic difficulties, has been supported by some leading nations and world organizations (European Commission, 2010a, 2010b; National Information Infrastructure Advisory Council, 1996; OECD, 2011; Unesco, 2009; US Department of Commerce, 2000, 2002). At the World Summit on the Information Society (WSIS), sponsored by the United Nations (UN), it was declared that the global challenge for the new millennium is to build a society “where everyone can create, access, utilize and share information and knowledge, enabling individuals, communities and peoples to achieve their full potential in promoting their sustainable development and improving their quality of life” (WSIS, 2003, 2005).

In this context, the existence of the digital inequalities both

between and within countries, poses a major threat to the fulfilment of ICT potential. The digital divide has been defined as “the gap between individuals, households, businesses and geographic areas at different socio-economic levels with regard both to their opportunities to access ICT and to their use of the Internet for a wide variety of activities.” (OECD, 2001).

There have been multiple efforts to quantify/measure the international digital divide, that is, the digital gap across countries (Cruz-Jesus, Oliveira, & Bacao, 2012; Cuervo & Menéndez, 2006; Dewan, Ganley, & Kraemer, 2005). However, these efforts have generally neglected the fact that within each country there might also be digital inequalities related to population's socio-economic imbalances. Research on digital divide has shown that several socio-economic factors lead to asymmetries in ICT adoption and use between individuals: income, age, and educational attainment, among others (Dewan & Riggins, 2005). Education, in particular, reveals itself as an extremely important factor because, not only are more educated individuals more likely to have less difficulty coping with technology's complexity (Rogers, 2005), but they will also most likely be exposed to ICT in their professional and personal lives. This paper focuses on measuring the education-related digital divide, i.e., the digital gap between countries, controlling for the level of education of their population, which is to say, the digital asymmetries that may exist in each country due to differences in education level of the country's population.

The context of our study is the European Union (EU), which is

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not immune to the digital divide, and recognizes that developing a digital economy based on knowledge and innovation is a critical issue for the Union's present and future growth and competitiveness (European Commission, 2010b). The European Commission's (EC) 2020 Strategy seeks "a smart, sustainable and inclusive growth for European Economy" (European Commission, 2010b) and "to exit the crisis and prepare the EU economy for the challenges of the next decade" (European Commission, 2010a). The Digital Agenda for Europe, included in the Europe 2020 Strategy as the first of the seven strategy's flagships, aims to reboot Europe's economy and help Europe's citizens and businesses to get the most out of ICT (European Commission, 2010a). Hence, detecting and correcting digital inequalities becomes a must in order to avoid jeopardizing 2020 Strategy's objectives.

The goal of this paper is to address the role that educational imbalances have on the digital divide among the citizens of the 28 member states of the European Union (EU-28). In particular, the research questions of this paper are the following: (1) what are the most important features of the digital divide across the 28 member states of the European Union? (2) To what extent is there an educational-related domestic digital divide in the EU-28 and how does it shape countries' imbalances? In answering these questions the remainder of the paper is organized as follows: Section 2 presents the theoretical background; Section 3 describes the data; Section 4 presents the methodology, analysis, and discussion of the results; Section 5 draws some concluding remarks.

2. Theoretical background

2.1. Literature review

Recent decades have witnessed the emergence of ICT as, perhaps, the key general purpose technology (GPT – technological innovations that have the potential to improve most industries and society sectors) of present times (Bresnahan & Trajtenberg, 1995; Doong & Ho, 2012; European Commission, 2013). This "digital" revolution began with the automation and computerization of manufacturing, and was followed by the widespread of personal computers (PC) and the Internet, which led to the fact that broad sectors of the economy, previously untouched by ICT, benefited from these through investment and productivity improvements. Tertiary (market services) sectors, which account for the major portion of Gross Domestic Product (GDP) in developed economies, have experienced considerable benefits from these technologies. Thereafter, non-market sectors such as financial, health, education, and even government-services have become more prone to the positive growth effects from ICT (European Commission, 2013).

ICT in general and the Internet in particular have allowed the implementation of a whole range of new services that have completely changed the way individuals and firms interact and communicate, do business, pursue economic growth, improve welfare, and even the way politics are conducted (OECD, 2004; Zhao, Kim, Suh, & Du, 2007). Internet browsing, email, VoIP, blogs, multimedia online streaming, social networking, on-line job seeking, wiki-sites, access to online libraries, e-commerce, and services like e-government, e-health, e-learning, and e-banking are examples of new possibilities that allow new types of communications and interactions for individuals, firms, and governments (Çilan, Bolat, & Coskun, 2009; European Commission, 2006; Facer, 2007; Forman, 2005; Hajli, 2014; Mutula & Brakel, 2006; Niehaves & Plattfaut, 2013; Vicente & Gil-de-Bernabé, 2010; Vicente & Lopez, 2010b).

Despite the multiple benefits that ICT may bring, not everything regarding these technologies does necessarily leads to positive outcomes. The idea that ICT will enable a whole new world of

endless opportunities, liberated from problematic sociocultural aspects, such as gender, age, race, and geography is utopic (Gunkel, 2003). As the spread of ICT became wider, it quickly became clear that access (and later on, use) was limited by specific constraints and should not be assumed by researchers and policy makers to be either universal or instantaneous. It was within this context that the term "digital divide" appeared. Although the literature and forums on the subject regularly attribute the term to Larry Irving Junior, former Assistant Secretary for Communications and Information of the US Department of Commerce, the fact is that it was not authored by him, as he himself admitted years later (Gunkel, 2003). The "digital divide" term became popular in the third "Falling Through the Net" report, from the US Department of Commerce's National Telecommunications and Information Administration (NTIA) (US Department of Commerce, 1999), which defined it as "the divide between those with access to new technologies and those without". Within the series of these reports (US Department of Commerce, 1995, 1998, 1999, 2000, 2002) the definition of digital divide evolved from merely PC ownership, to the inclusion of Internet access, and later, to the availability of broadband connections and the types of online usages. In fact, literature distinguishes between the inequalities about access to and use of ICT, the so-called first- and second-order digital divides first- and second-order digital divides (Dewan & Riggins, 2005; DiMaggio, Hargittai, Celeste, & Shafer, 2004). In the first-order digital divide the inequalities are with regard to ICT access, while in the second-order the problem is postulated in terms of different use patterns and intensity among individuals/organizations that already have (very similar) access to ICT (e.g., using the Internet just for web-browsing or email vs. using it for e-learning, social-network, applying to jobs online, e-banking, e-health, etc.).

Research has shown that both types of divides are mainly shaped by socio-economic inequalities among countries and individuals. Hence, those who are economically and sociologically disadvantaged (i.e., individuals with lower incomes or education levels, those with disabilities, living in rural areas, belonging to ethnic minorities, women, and the elderly) are more likely to suffer from digital inequalities (Azari & Pick, 2005; Crenshaw & Robison, 2006; Ferro, Helbig, & Gil-Garcia, 2011; Hilbert, 2011; Hill, Betts, & Gardner, 2015; Lengsfeld, 2011; Novo-Corti, Varela-Candamio, & García-Álvarez, 2014; Payton, 2003; Vicente & Lopez, 2006, 2008, 2010b, 2010c; Wu, Chen, Yeh, Wang, & Chang, 2014). In addition to whom the problem of digital divide may affect, it may also be related to the type of digital divide, i.e., access or use of ICT – which may appear to be the same, but are not (e.g., Goldfarb & Prince, 2008; Orviska & Hudson, 2009). Hsieh, Rai, and Keil (2008, 2011) showed noticeable differences in ICT access versus use patterns between those who are economically advantaged and disadvantaged. Besides the intrinsic motivations for access to ICT, these individuals also have very different post-implementation behavior regarding the use of these technologies. Hsieh et al. concluded that economically advantaged people have a "higher tendency to respond to network exposure", using ICT with much more confidence than the disadvantaged. Novo-Corti and Barreiro-Gen (2015) studied the ICT adoption in households of different Spanish regions, finding that besides ICT infrastructure, households' characteristics such as income, education, and age of its individuals, affect ICT adoption. Wu et al. (2014) addressed the digital divide between children with and without learning disabilities in Taiwan, and although he concluded that there was no divide in ICT access, a different conclusion was drawn for ICT use. In other words, "there was a significant difference in ICT competencies between children with and without learning disabilities". In addition to these socio-demographic characteristics of individuals, Venkatesh and Sykes (2013) found that social network aspects are also significant in

explaining ICT use, and even value. As noted by Epstein, Nisbet, and Gillespie (2011), different types of inequalities (first- and second-orders) require different actions: Inequalities in ICT access may be bridged, for example, through the subsidization of these technologies, since they are mainly due to economic constraints, which leads to lack of conditions to have access to ICT; whereas for inequalities in ICT use (not access) dependent more on lack of skills and awareness toward ICT, in which education and training appears as a critical factor.

2.2. Education as a driver of ICT

Inequalities in education attainment of individuals are one of most mentioned reasons for the digital divide (Cruz-Jesus et al., 2012; Kiiski & Pohjola, 2002; Shirazi, Ngwenyama, & Morawczynski, 2010). The theoretical foundations for this cause–effect relationship are multiple and can be found in several different theories. The diffusion of innovations theory (DOI), claims that complexity is a major obstacle for technology's adopters (Rogers, 2005). Hence, the easier a technology is to adopt, the faster is its adoption rate (Katz & Aspden, 1997). This fact makes of individuals' educational features to play a critical role, as when technically challenged, more educated individuals are more likely to be more prone to flexibly and effectively overcome complexity's constraints (Zhao et al., 2007). Thus, as in the case of ICT, when one is interacting with it, the relatively higher educational attainment should make it easier to cope with the complexity of the technology, minimizing the impact of the difficulties (Hsieh et al., 2008). In this sense education facilitates the absorption and comprehension of information, leading to an increasing information divide between higher- and lower-educated individuals. This is, in fact, the main argument lying behind the knowledge gap theory (Tichenor, Donohue, & Olien, 1970), developed within the context of the mass media's (TV, radios, etc.) wide dispersion. Tichenor et al. (1970) posit that, “as the infusion of mass media information into a social system increases, higher socioeconomic status segments tend to acquire this information faster than lower socioeconomic status population segments so that the gap in knowledge between the two tends to increase rather than decrease”. If this is so in mass media technologies, which are far less complex and challenging than the Internet-related ones, then in the case of these last, that fact should be even more noticeable. Mass media technologies are not as demanding as are ICT, since they do not require a set of actions by its users. ICT and Internet related activities, however, require that the users navigate throughout a great amount of information instead of being (almost) as mere receptors of information (Bonfadelli, 2002). Moreover, in case of ICT, although access is a prerequisite, it is not enough, *per se*, for taking all the advantages these technologies can bring, as important differences may remain in the nature of Internet use (van Dijk, 2005). As Vicente and Lopez (2006) point out “not only does the user need access to infrastructure but also he needs the ability to access to information, to look for it and to use it”. Moreover, it is also reasonable to hypothesize that more educated individuals are more likely to work in information-intensive industries, thus using ICT more often and intensively whether at work or at home. Accordingly, Howard, Rainie, and Jones (2001) found that more educated individuals will tend to employ the Internet more productively and to greater economic gain than those with lower educational attainments, consistent with the findings of Hargittai and Hinnant (2008). More recently Peng, Fan, and Dey (2011) found that individuals who use PC at work or school are more likely to adopt ICT. A consistent finding has been demonstrated by Tengtrakul and Peha (2013), in which authors conclude that “the higher the educational level of students, the stronger the increase in likelihood of a household adopting ICT”.

3. Data

Consistent with our theoretical framework, in which ICT are classified as GPT, we measured their use across the citizens of the EU-28. The indicators were selected based on the literature and the availability of educational breakdowns for them. The variables were extracted from the Digital Agenda Scoreboard database – all pertaining to the year 2013. The fact that all variables were obtained from the official Digital Agenda Database (Digital Agenda key indicators) guarantees that the results of the analysis enjoy a high degree of reliability. Table 1 shows the variables collected and makes reference to other earlier research that used these variables to measure the digital divides.

All these indicators refer to different types of usages, and allow us to assess the differences not only across countries, but also within different educational levels. As our study focuses on measuring the digital adoption of individuals, we make our best efforts to include indicators that measure the ICT pervasiveness at multiple educational levels (low, medium, and high) for individuals. Our rationale is the following. Web-browsing and using the Internet to search for information about goods and services are some of the most general activities individuals use the Internet for; therefore, the percentage of individuals regularly using the Internet (IntPop) and looking for information about goods and services online (IntSrc) is an effective measure for assessing the use of ICT of individuals for general purposes (Cruz-Jesus et al., 2012). The recent emergence of the mobile Internet access is also an important feature of the information society (Okazaki, 2006). For this reason, we find it especially interesting to measure the percentage of population using mobile devices to access the Internet (Mob). Considering that we are attempting to analyze the digital divide at a European level, it is also important to consider the position of the European entities concerning the digital divide. Via the Digital Agenda for Europe (European Commission, 2010a), the European Commission, emphasizes the role of specific electronic services, more specifically e-health, e-learning, e-banking, and e-government. E-banking and e-health are considered to be “some of the most innovative and advanced online services” (European Commission, 2010a); e-government services are also highlighted in the Digital Agenda, since “despite a high level of availability of e-government services in Europe, differences still exist amongst Member States” (European Commission, 2010a). The inclusion of these indicators, related to how the use of these advanced services also allows us to analyze the so called “second-order digital divide”, expands our focus from mere ICT adoption, to include the manner in which it is used by individuals (Hsieh et al., 2008). For these reasons, the percentage of individuals using e-health, e-learning, e-banking, and e-government services (eHealth, eLearn, eBank, and eGov, respectively) were included in our analysis.

Moreover, ICT allows that “an increasing number of people engage themselves and others in e-participation, that is, in social and political participation” (Vicente & Novo, 2014), as demonstrated in the popular example of the 2008 US Primary Presidential Campaign (Wattal et al., 2010). ICT are becoming a critical tool not only for politics, but also for individuals to engage in public and political activities, serving as a bridge between political entities and citizens. Interestingly, as Vicente and Novo (2014) posit, there are mixed conclusions when it comes to the causes of asymmetries in e-participation, as some authors find no evidence on the effect individuals' education over e-participation, while others do. Nevertheless, the use of ICT for civic participation (eCivic) allows individuals to get information on public issues in a faster and easier way than with traditional (offline) methods, which ultimately facilitates political and civic engagements and minimizes social

Table 1
Description of variables and earlier evidence on their importance used.

Code	Variable	Support
IntPop	Individuals regularly using the Internet (every day or almost every day)	(Billon, Ezcurra, & Lera-López, 2008; Billon, Marco, & Lera-Lopez, 2009; Çilan et al., 2009; Cruz-Jesus et al., 2012; Vicente & Lopez, 2010a)
IntSrc	Individuals looking for information about goods and services online	(Cruz-Jesus et al., 2012; Cuervo & Menéndez, 2006)
Mobile	Individuals using a mobile device to access the internet, away from home or work	(International Telecommunication Union, 2013; Kauffman & Techatassanasoontorn, 2005; Okazaki, 2006; Thompson & Garbacz, 2011; Vicente & Lopez, 2006; Vicente & Lopez, 2008)
eHealth	Individuals seeking online information about health	(Cruz-Jesus et al., 2012; European Commission, 2010a)
eLearn	Individuals looking online for information about education, training or course offers	(Çilan et al., 2009; Cruz-Jesus et al., 2012; European Commission, 2010a)
eBank	Individuals using online banking	(Cruz-Jesus et al., 2012; European Commission, 2010a)
eCivic	Individuals taking part in online consultations or voting to define civic or political issues	(Epstein, Newhart, & Vernon, 2014; Helbig, Ramón Gil-García, & Ferro, 2009; Vicente & Novo, 2014; Warren, Sulaiman, & Jaafar, 2014; Wattal, Schuff, Mandviwalla, & Williams, 2010)
eGov	Individuals interacting online with public authorities, last 12 months	(Çilan et al., 2009; Cruz-Jesus et al., 2012; European Commission, 2010a)
eCom	Individuals ordering goods or services online	(Cruz-Jesus et al., 2012; Cuervo & Menéndez, 2006; European Commission, 2010a; Turban, King, Lee, Liang, & Turban, 2015)
eCom_CB	Individuals ordering goods or services online, from sellers from other EU countries	(European Commission, 2010a, 2013; Turban et al., 2015)

constraints (Anduiza, Gallego, & Cantijoch, 2010). For this reason, eCivic measure was included in our framework. Finally, in our analysis, we also pay special attention to e-commerce (eCom), considering the importance of it in the global economy (Turban et al., 2015). The European Commission recently drew four scenarios for the role of ICT in European economic growth, and the one with the highest expected growth rate (average 2.5% GDP per year) is the one in which EU nation-based ventures go across borders and compete globally. If this is accomplished, it is expected that 60% of the EU economic growth (around 1.5% a year) is due to ICT contribution. It is estimated that in 2010 92% of Europeans who ordered goods or services over the internet did so from national sellers, rather than cross-border ones (European Commission, 2010a). For these reasons, cross-border e-commerce (eCom_CB) was included in our set of variables.

Data on these 10 variables are available by educational breakdown. In particular, three levels of education are available and were considered: no formal, primary or lower secondary education (ISCED 0, 1 or 2); Upper or post-secondary education, but not tertiary (ISCED 3 or 4); and Tertiary education (ISCED 5 or 6). Each of these classifications are named low, medium, and high education class, respectively, hereafter.

The data (see Table 2) show noticeable imbalances in ICT adoption and use of the individuals belonging to the 28 European countries: in Romania only 45% of the population uses the Internet regularly, while in Luxembourg 93% do so, more than twice. These same two countries are also at different ends of the spectrum in the use of mobile devices to access the Internet (Mobile), with values of 7% and 43%, respectively. In the use of e-banking services, the percentage of individuals using these services ranges from 4% in Romania up to 84% in Finland, a value 21 times higher. The higher asymmetries are present in the percentage of individuals using cross-border e-Commerce, as in Romania only 1% do so whereas in Finland this value figure is 64%. We also notice extreme asymmetries in the overall profile of the 28 European countries. Luxembourg is the best-ranked country in 4 of the 10 variables used, followed by Finland with 3, while Romania is the poorest-ranked with 8. These uneven distributions can tell us a great deal about the asymmetries that exist between countries. Nevertheless, the dimensionality of the data used - 10 variables for 28 countries - makes it impossible to address the digital divide with simple descriptive statistics. This is why the use of multivariate statistical methods is more appropriate for analyzing all these digital asymmetries.

4. Analysis

4.1. Methodology: factor analysis

Factor analysis is a multivariate technique that focuses on examining the underlying patterns of complex and multidimensional phenomena and allows to summarizing it in a small set of factors, thereby making its interpretation easier (Hair, 2014). This technique is especially suitable for the analysis of the digital divide, given the pervasiveness of ICT across the full society and economy (Cruz-Jesus et al., 2012) and its wide set of applications (e-health, e-banking, e-learning, etc.), which require taking into account multiple dimensions.

The application of this technique is in several steps (Sharma, 1996). The basis for its use is that data present an underlying correlation structure. If so, then it is possible to proceed with the analysis and obtain the factors. Once these have been calculated, the next step is to select the least number of factors that facilitate the interpretation of the phenomenon under analysis, while keeping the maximum of the original information. Finally, the selected factors are interpreted according to the factor loadings, i.e., correlation with original (contributing) variables.

The next two subsections describe the factor analysis performed on the ICT variables. The factor analysis is first carried out using aggregated data at country level for each of the member states. Second, data are disaggregated by educational level in order to assess the internal digital divides. The first procedure allows us to assess the digital divide across the member states of the European Union. By comparing its results with those of the second, it is possible to assess the extent to which an aggregate analysis can hide important insights about the digital divide. That is, while a country might be on average well positioned, there might be groups in its population suffering from digital exclusion. Only by running these two analyses simultaneously is it possible to expose the partial nature (if such is the case) of the picture provided by an analysis performed only at the country-level, which is the standard practice in the literature.

4.2. The digital divide across EU-28

As mentioned above, the success of factor analysis depends on the correlation structure in the input data (Hair, Anderson, Tatham, & Black, 1995). We therefore need to confirm that this correlation exists, otherwise the factor analysis may provide weak results.

Table 2
Data and descriptive statistics of the variables used for measuring the digital development.

Country	IntPop	IntSrc	Mobile	eHealth	eLearn	eBank	eCivic	eGov	eCom	eCom_CB
Austria	0.77	0.67	0.26	0.49	0.35	0.49	0.11	0.54	0.54	0.39
Belgium	0.80	0.67	0.24	0.43	0.20	0.58	0.04	0.50	0.48	0.28
Bulgaria	0.51	0.35	0.14	0.27	0.22	0.05	0.03	0.23	0.12	0.06
Croatia	0.63	0.46	0.22	0.35	0.37	0.23	0.07	0.25	0.26	0.07
Cyprus	0.62	0.56	0.15	0.32	0.19	0.23	0.03	0.30	0.25	0.21
Czech Rep.	0.70	0.63	0.18	0.41	0.13	0.42	0.03	0.29	0.36	0.07
Denmark	0.91	0.82	0.35	0.54	0.41	0.83	0.12	0.85	0.77	0.32
Estonia	0.77	0.60	0.33	0.39	0.33	0.73	0.06	0.48	0.23	0.13
Finland	0.89	0.79	0.37	0.60	0.32	0.84	0.19	0.69	0.65	0.32
France	0.78	0.68	0.27	0.49	0.31	0.58	0.10	0.60	0.59	0.18
Germany	0.80	0.77	0.27	0.58	0.36	0.47	0.11	0.49	0.69	0.11
Greece	0.56	0.50	0.13	0.34	0.21	0.11	0.05	0.36	0.25	0.09
Hungary	0.71	0.58	0.16	0.49	0.34	0.27	0.03	0.37	0.28	0.06
Ireland	0.75	0.61	0.39	0.38	0.33	0.46	0.04	0.45	0.46	0.24
Italy	0.56	0.37	0.13	0.32	0.26	0.22	0.06	0.21	0.20	0.07
Latvia	0.71	0.52	0.16	0.37	0.31	0.55	0.04	0.35	0.32	0.15
Lithuania	0.65	0.57	0.09	0.41	0.24	0.46	0.06	0.34	0.26	0.10
Luxembourg	0.93	0.70	0.43	0.58	0.48	0.63	0.13	0.56	0.70	0.64
Malta	0.66	0.50	0.28	0.41	0.42	0.43	0.09	0.32	0.46	0.39
Netherlands	0.92	0.83	0.36	0.57	0.29	0.82	0.07	0.79	0.69	0.15
Poland	0.60	0.46	0.20	0.27	0.18	0.32	0.02	0.23	0.32	0.03
Portugal	0.58	0.46	0.19	0.42	0.35	0.23	0.07	0.38	0.25	0.11
Romania	0.45	0.26	0.07	0.27	0.22	0.04	0.02	0.05	0.08	0.01
Slovakia	0.74	0.56	0.27	0.44	0.29	0.39	0.03	0.33	0.45	0.17
Slovenia	0.69	0.55	0.19	0.50	0.32	0.32	0.08	0.52	0.36	0.15
Spain	0.66	0.48	0.23	0.44	0.46	0.33	0.10	0.44	0.32	0.11
Sweden	0.92	0.81	0.36	0.56	0.35	0.82	0.14	0.78	0.73	0.23
U.K.	0.87	0.70	0.35	0.45	0.34	0.54	0.08	0.41	0.77	0.14
Minimum	0.45	0.26	0.07	0.27	0.13	0.04	0.02	0.05	0.08	0.01
Maximum	0.93	0.83	0.43	0.60	0.48	0.84	0.19	0.85	0.77	0.64
Average	0.72	0.59	0.24	0.43	0.31	0.44	0.07	0.43	0.42	0.18
Std. Dev.	0.13	0.15	0.10	0.10	0.09	0.23	0.04	0.19	0.21	0.14

The correlation matrix (see Table 3) shows that each variable has at least one absolute correlation coefficient of 0.60 with another variable. Although this correlation value is not extremely high, it ensures that all variables are measuring the same phenomenon – ICT adoption of individuals. We notice that some pairs of variables have very high correlation levels. The percentage of population regularly using the Internet (IntPop) has a correlation of 0.94 with the percentage of individuals using the Internet for finding commercial information about products or services (IntSrc) and a value of 0.92 with the percentage of individuals using e-banking services (eBank). At the other end of the spectrum we have the percentage of individuals looking online for information about education, training or course offers (eLearn) with a correlation level of 0.30 with the percentage of individuals using the Internet for finding commercial information about products or services (IntSrc).

Some more insights on the correlations of the data can be assessed by means of the Kaiser-Meyer-Olkin measure (KMO), in which values over than 0.5 indicate that the correlation is enough

as to proceed with factor analysis. In the present analysis, the value of the KMO measure was 0.85, which expresses a very good suitability (Jolliffe, 2005; Sharma, 1996).

In order to obtain the factors, we used the standard factor analysis, which is widely used in Marketing and the Social Sciences (Peres-Neto, Jackson, & Somers, 2005). Then, we have to decide how many factors are to be extracted from the analysis in order to facilitate the understanding of the phenomenon, while keeping as much information as possible. There are several criteria on the number of factors to extract (Sharma, 1996): Pearson's (variance explained above 80%); Kaiser's (eigenvalues above 1); and the Scree Plot (visually interpretation of the solutions, selecting the one immediately before a breaking point). All of these methods were considered and yielded the same solution: the optimal number of factors to be extracted is two. As shown in Table 4, the percentage of variance retained by these two factors is 84%. To measure the scale reliability of each factor, Cronbach's alpha was also calculated. It measures the internal consistency of each factor within itself. The

Table 3
Correlation matrix.

	IntPop	IntSrc	Mobile	eHealth	eLearn	eBank	eCivic	eGov	eCom	eCom_CB
IntPop	1	0.94 **	0.86 **	0.86 **	0.45 *	0.92 **	0.62 **	0.87 **	0.91 **	0.61 **
IntSrc		1	0.74 **	0.86 **	0.30	0.87 **	0.60 **	0.89 **	0.89 **	0.50 **
Mobile			1	0.68 **	0.58 **	0.78 **	0.61 **	0.74 **	0.82 **	0.67 **
eHealth				1	0.56 *	0.75 **	0.75 **	0.85 **	0.83 **	0.55 **
eLearn					1	0.36	0.63 **	0.44 *	0.46 *	0.53 **
eBank						1	0.62 **	0.86 **	0.79 **	0.53 **
eCivic							1	0.70 **	0.67 **	0.59 **
eGov								1	0.81 **	0.53 **
eCom									1	0.60 **
eCom_CB										1

* – Correlation is significant at the 0.05 level (2-tailed); ** – Correlation is significant at the 0.01 level (2-tailed).

Table 4

Results of factor analysis (Factor loadings, variance explained, and Cronbach's Alpha).

Rotated factor model: Varimax		
	Factor 1	Factor 2
IntSrc	0.96	0.19
IntPop	0.92	0.34
eBank	0.89	0.26
eGov	0.86	0.34
eCom	0.85	0.39
eHealth	0.77	0.48
Mobile	0.71	0.49
eLearn	0.12	0.92
eCom_CB	0.38	0.71
eCivic	0.48	0.70
Variance (%)	55%	28%
Variance Total	55%	84%
Cronbach's Alpha	0.96	0.70

values returned were 0.96 for factor 1 and 0.70 for factor 2, which confirm the high scale reliability of the two factors extracted (Nunnally, 1978).

The next step is to provide meaning to the two factors identified. The rotation of factors makes this task much easier as it allows a better split of the original variables between the factors (Sharma, 1996). Although there are several types of rotation, including orthogonal and oblique methods, the orthogonal ones seem to be the most widely used (Sharma, 1996). Varimax rotation is applied in the case of our study. We observe that out of the 10 original variables, seven of them are highly correlated (correlation over 0.7) with the first factor, while the remaining three have large correlations with second factor. By paying attention to the variables correlated with each factor, it is possible to interpret the meaning of them. The first dimension is related to the widespread of ICT among individuals, and could be labelled the *general ICT adoption by individuals*. This dimension includes the Internet use for a wide variety of activities, ranging from simple browsing to the use of specific advanced services, and the usage of mobile devices to access them. The second dimension is labelled *eLearning, cross-border eCom, and civic participation* since it is highly correlated with the take up of these services by Europeans.

In order to assess the performance of each country in each dimension, factor scores can be computed. These scores are obtained as the product between the standardized factor score coefficients and the corresponding standardized values of the original set of variables.

Fig. 1 shows countries' scores in the two factors/dimensions identified. The first factor, *general ICT adoption by individuals*, is plotted in the horizontal axis, where the right side indicates that a country scores higher than the European average and correspondingly, the left side shows that it is below it. The second factor, *eLearning, cross-border eCom, and civic participation*, is represented on the vertical axis, where the top and bottom sides mean that a country is over or beneath the mean, respectively. Hence, the upper-right quadrant includes those countries that score higher than the European average on both dimensions (+/+). The lower-left quadrant shows just the opposite situation, countries below the mean in both dimensions (−/−).

As one can see, the best-ranked countries in the first dimension, *general ICT adoption by individuals*, are the Netherlands, Sweden, Denmark, and Finland, while Romania and Bulgaria present the lowest levels in this dimension. In the second dimension *eLearning, cross-border eCom, and civic participation*, Luxembourg is far ahead among the member states, followed by Malta and Spain, while the Czech Republic and Poland are at the other end of the spectrum. Looking at upper-right and lower-left quadrants, it is

possible to identify the best and worst performers in the two dimensions, respectively. The first group is composed of Northern and Central European Countries, while the second one includes Western and Southern states. This confirms the idea that the digital divide mirrors, at least to some extent, the traditional socio-economic divides between nations, a finding in line with earlier research (e.g., Çilan et al., 2009; Cruz-Jesus et al., 2012). Among the countries in the other quadrants, it is interesting to note that Dutch position: while it is the best-positioned country in the first dimension, it scores negatively in the second one. A similar case is that of its geographic neighbour, Belgium, which also scores negatively in the second dimension, while showing a positive score in the first, although not as good as the Dutch. Both countries are thus lagging behind regarding the use of eLearning, cross-border eCom, and civic participation. The countries closest to the European average are Ireland, Estonia, Slovakia, and Slovenia.

4.3. The education-related digital divide across European member states

As the global position of European member states in the information society has been analyzed, we turn our attention to whether this position changes when educational attainment is taken into account, and if so, in what manner.

With that aim, education-related factor scores are obtained for each country. These scores are computed by multiplying the previously obtained factor score coefficients and the standardized values of the ICT variables, by the averages and standard deviations of the aggregated data, “drilled-down” by the educational classes specified above. With this approach we assure that both the aggregated and disaggregated countries are represented in the same space. The units of analysis are 84, i.e., the 28 European member states multiplied by the three educational classes (low, medium and high).

Fig. 2 shows countries' positions in the two dimensions once education has been taken into account. The meaning of the axes is the same as in Fig. 1. The quadrant in the top-right corner comprises those educational groups that are above the European average in both dimensions; conversely, the quadrant in the lower-left includes those that are below the mean. As one can see, the highly educated individuals of most member states are situated in the upper-right quadrant, with the exceptions of Malta, Spain, Italy, and Romania, which are in the upper-left quadrant; and Poland and the Czech Republic in the lower-right quadrant. Note that, therefore, no higher educated nationals are in the lower-left quadrant, the one with the lowest levels on both dimensions. It is also worth noticing that the Finnish population, regardless of its education level, is above the European average in the two dimensions of ICT use. The upper-right quadrant also shows that, exceptionally, the low-educated populations of some Northern countries are using ICT more than the average, those being the less educated Swedish and Danish. The same happens with the Austrians, with medium educational attainment. The remaining low-educated Europeans are below the European average in the two identified dimensions and are situated in the lower-left corner, except for those in Luxembourg, who show a negative score in the first dimension and positive in the second, and the Dutch with just the opposite situation, positive score in the first dimension and negative in the second. Medium-educated Europeans are scattered over the two left quadrants and the lower-right quadrant.

Comparing Figs. 1 and 2 leads to some interesting insights. For example, when aggregated figures are considered, factor analysis shows that Latvia is positioned in the lower-left quadrant, meaning that it is below the European average in both dimensions (Fig. 1). However, when ICT figures are disaggregated by educational level,

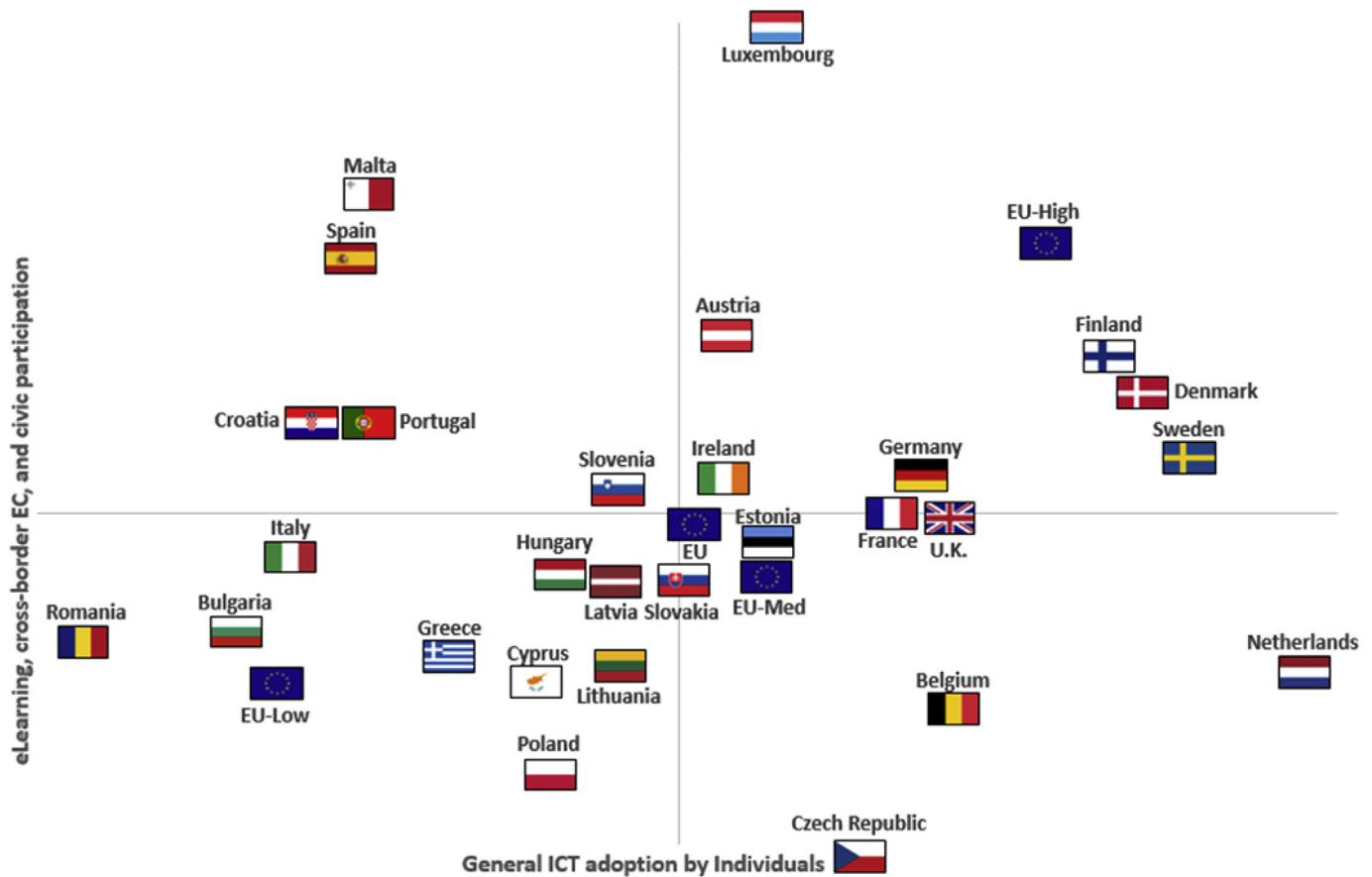


Fig. 1. Countries' projections.

we see that Latvians with higher education score positively in both dimensions compared to the rest of educational groups, which score negatively (Fig. 2). In the case of the United Kingdom, the aggregated results show that it is above the average regarding *general ICT adoption by individuals*, whereas it is below average for eLearning, cross-border eCom, and civic participation (Fig. 1). By educational level, the United Kingdom shows some important gaps: the higher educated British perform positively in both dimensions, in contrast to the lower educated who score negatively and those with medium level education, who are good in terms of adoption but still lag behind in the take up of some online activities, i.e., e-learning, e-commerce, and e-participation (Fig. 2).

Table 5 summarizes the results of each country on each dimension when the analysis is performed with the population taken as a whole and when figures are disaggregated by educational level. This table clearly shows how the analyses performed on aggregated data can mask important internal divides within countries. In addition to the above-mentioned cases of Latvia and the United Kingdom, it is also worth mentioning what happens with Austria, Ireland, and Luxembourg. These three countries score positively on an aggregated level, meaning that their populations are above the European average in the two dimensions identified. However, when educational level is taken into account, it is possible to observe some problems with the low-educated Luxembourg population (negative score in the first dimension) and Austrians (negative score in both dimensions) and with the low and medium-educated Irish (the former with negative scores in both dimensions and the latter scoring negatively in the second dimension).

In order to gain some more insights on the internal digital divides described by education, we have measured the distances between factor scores using two different perspectives. In both, these distances correspond to the definition of the Euclidean distance between two points, where the scores in the first and second dimensions define the first and second elements of the coordinate, respectively. The first perspective involves calculating the distance between nationals of each of the three education classes to the national average (see Fig. 3), and the second is calculating the combination of distances between the three (Fig. 4). These two approaches are thus complementary.

When analyzing the differences between each educational class with the respective country, as seen in Fig. 3, not surprisingly the medium educated individuals are those who best represent the aggregated national average. Hence, the main differences lies between those with low and high educations. However, the latter seems to distance themselves in a more intensive manner than the former. In other words, the gap between the higher educated citizens, with respect to the national average, is stronger than the lower educated when it comes to ICT use in 22 of the 28 member states.

Concerning the relative position between education classes, it is clear that, naturally, the largest divides are between the highest and lowest educated people, with Malta, Portugal, and Spain showing the biggest gaps. These three countries also show the largest divides between the medium and low levels of education, although Portugal and Malta interchange their positions. Note that the distance from low to medium educated, and from medium to high educated people, in terms of digital development, is somehow

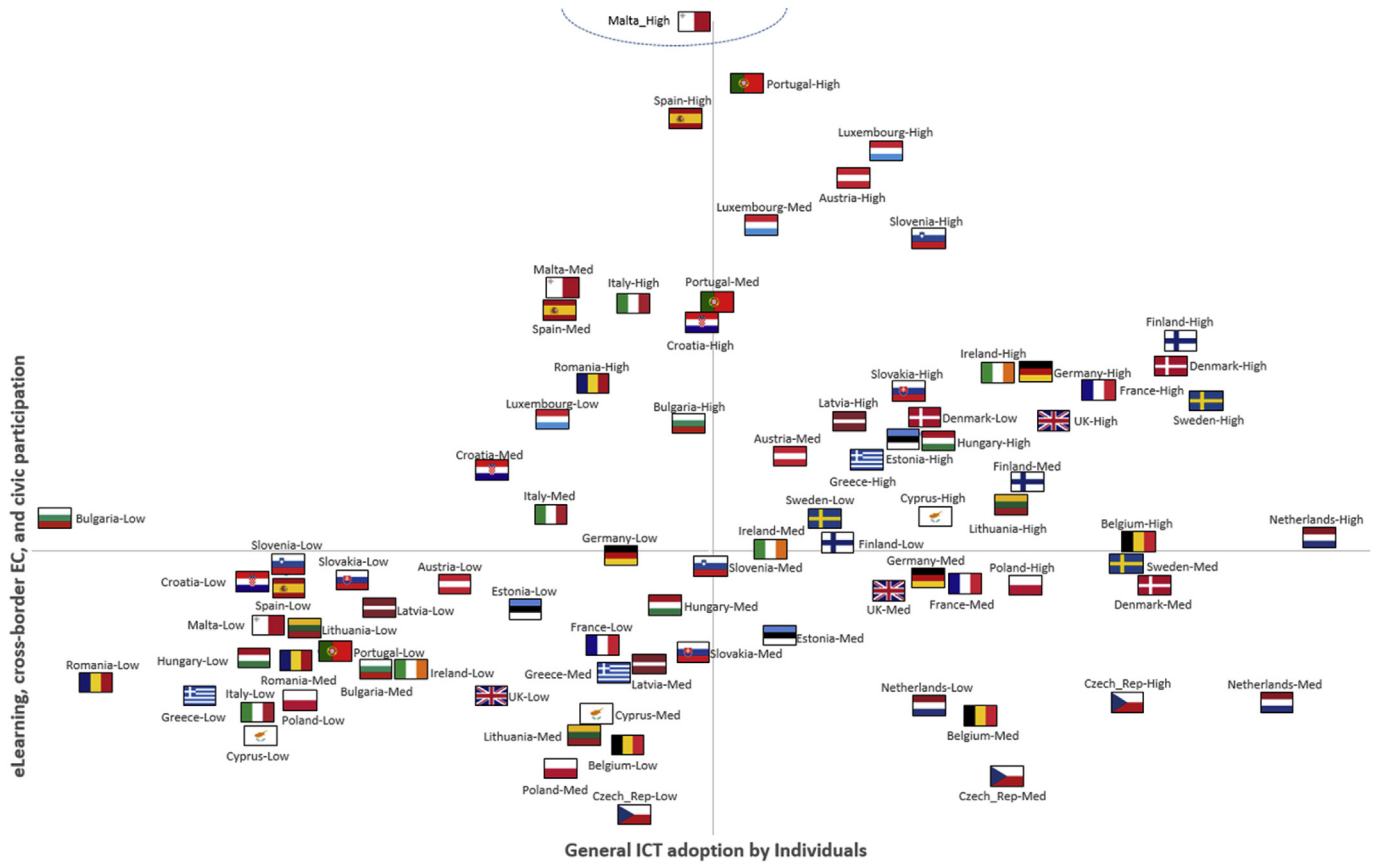


Fig. 2. Countries' projections by citizens' educational attainment.

Table 5

Countries' scores in each dimension.

Country/Education	Aggregated		Low		Medium		High	
	Factor 1	Factor 2	Factor 1	Factor 2	Factor 1	Factor 2	Factor 1	Factor 2
Austria	+	+	–	–	+	+	+	+
Belgium	+	–	–	–	+	–	+	+
Bulgaria	–	–	–	+	–	–	–	+
Croatia	–	+	–	–	–	+	–	+
Cyprus	–	–	–	–	–	–	+	+
Czech Republic	+	–	–	–	+	–	+	–
Denmark	+	+	+	+	+	–	+	+
Estonia	+	–	–	–	+	–	+	+
Finland	+	+	+	+	+	+	+	+
France	+	–	–	–	+	–	+	+
Germany	+	+	–	–	+	–	+	+
Greece	–	–	–	–	–	–	+	+
Hungary	–	–	–	–	–	–	+	+
Ireland	+	+	–	–	+	–	+	+
Italy	–	–	–	–	–	+	–	+
Latvia	–	–	–	–	–	–	+	+
Lithuania	–	–	–	–	–	–	+	+
Luxembourg	+	+	–	+	+	+	+	+
Malta	–	+	–	–	–	+	–	+
Netherlands	+	–	+	–	+	–	+	+
Poland	–	–	–	–	–	–	+	–
Portugal	–	+	–	–	+	+	+	+
Romania	–	–	–	–	–	–	–	+
Slovakia	+	–	–	–	–	–	+	+
Slovenia	–	+	–	–	+	–	+	+
Spain	–	+	–	–	–	+	–	+
Sweden	+	+	+	+	+	–	+	+
U.K.	+	–	–	–	+	–	+	+

Note: + indicates a positive score, that is, the country is above the European average in that dimension; conversely, - indicates a negative score, that is, the country is below the European average in that dimension.

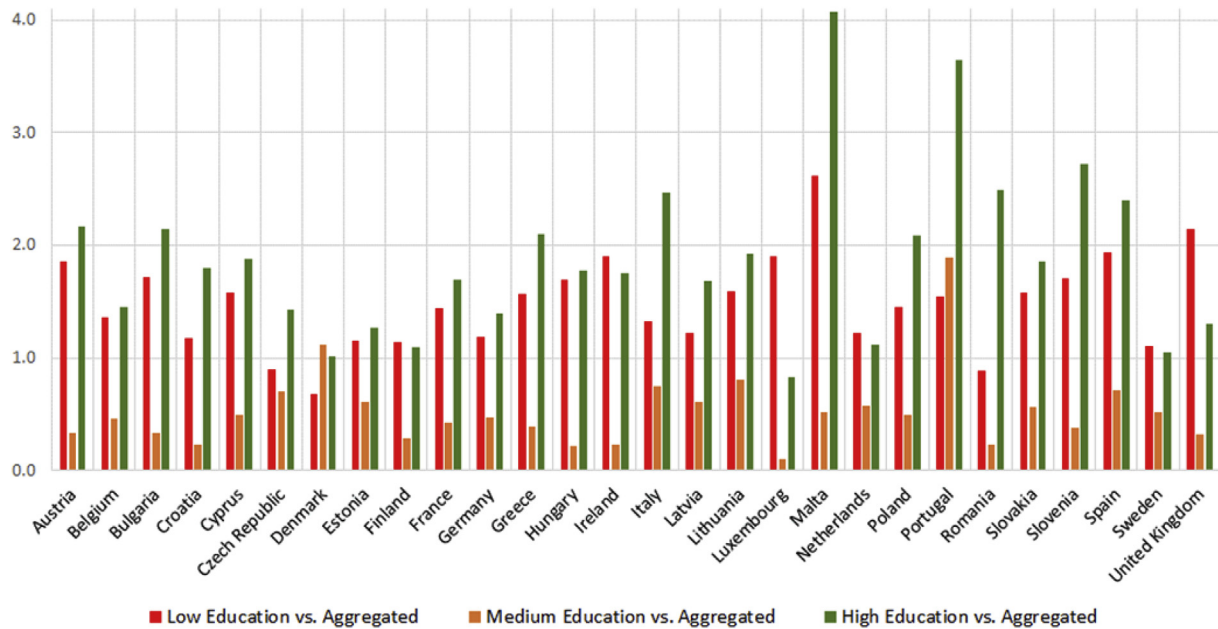


Fig. 3. The education-related digital divides – Education class vs. National average.

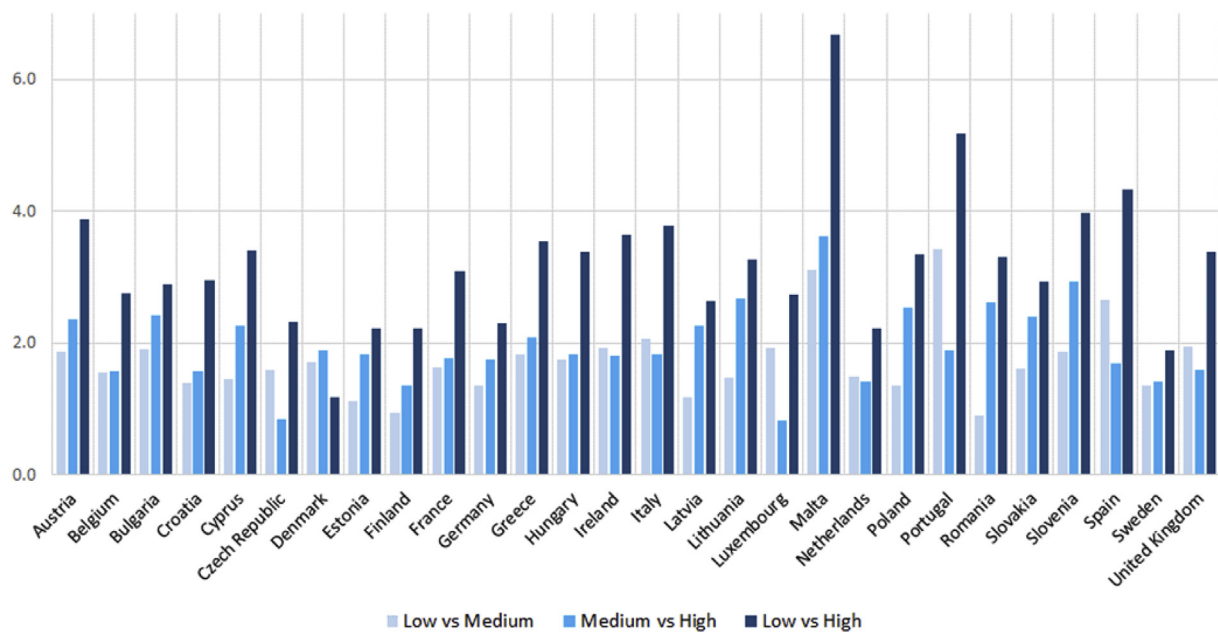


Fig. 4. The education-related digital divides – Education classes' distances.

similar. At the other end of the spectrum, Denmark and Sweden have the smallest digital divides between the highest and lowest educated. Regarding the divide between medium and low education levels, Finland and Romania show the smallest. Nonetheless, the situation of these two countries is radically different, since the former has very high levels of ICT penetration and use compared to the low levels of the latter. This figure also reveals a quite unexpected result: in Denmark it seems that there is a larger gap between medium and low educated people compared to the gap between the high and low educated. The reason for this result can be found in Fig. 2 and Table 5. They both show that while high and low educated Danish have positive scores on both dimensions, Danish with medium level of education score negatively in

dimension 2. Overall, the factor scores obtained by educational level and the calculation of the above-mentioned distances allow us to better assess the internal digital divides that exist within a country.

5. Concluding remarks

Our paper has analyzed the digital divide across and within the EU-28, combining as units of analysis, both aggregated countries and individuals according to their educational levels. Since most earlier studies have focused on identifying digital divides, whether across countries or, for a particular country, within its population, we consider our study as a necessary advance to the general

approach that takes account of the international digital divides but not of the internal gaps. Doing so is especially important since, as demonstrated, within a country some segments of the population might be lagging behind in the take up of ICT. Our analysis suggests that if we work only with aggregate levels, we might lose some important insights about the digital divide and fail to design appropriate policies as a result. Even in countries such as Finland, which is performing very well, there are some gaps to address within the population. In other countries such as Malta, Spain, and Portugal the big gaps are a concern. Hence, our analysis reveals that aggregated analysis can lead to false conclusions. Aggregated analysis shows only a part of the picture. This analysis should therefore complement traditional cross-country analysis.

Despite our best efforts to develop a complete and multidimensional analysis of the digital divide between and within countries of the European Union emphasizing the role of education asymmetries, some limitations must be acknowledged. First, our empirical application consists of just 10 variables, and, consequently, some features of the information society may very well not be covered by our data. Second, although we have analyzed the digital divide within EU countries, this analysis covered only educational imbalances between individuals, which implies that other domestic divide gaps were not covered, such as those regarding income or age of individuals. Finally, in third place, our analysis refers to the digital divide at a specific point in time, the year of 2013. Changes in this context are likely to occur with some speed, and our findings may become out-dated in the medium term.

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